

Fitts and GOMS

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tas: Marcello Bastea-Forte, Joel Brandt, Neil Patel, Leslie Wu, Mike Cammarano

09 October 2007

A little bit about this lecture

- http://www.youtube.com/watch?v=p5cPVP_llfo#

A little bit about this lecture

- Why is the Wii controller so much fun to use?



- Minimizing the distance between our human capabilities and what we want to the computer to do

A little about myself – Anoop Sinha

- Ph.D. ’03 UC Berkeley / B.S. ’96 Stanford
- Group-mate with Scott
- Did research on speech, pen, multimodal, multidevice user interfaces:
  - **Sinha’s Law**: the number of electronic devices each person uses regularly increases on average by +1 every year
- Worked in industry in Consulting and previously co-founded Danoo, which puts interactive digital screens in public places
- aks@cs.stanford.edu

Material from Stu Card's Lecture and James Landay's Lecture





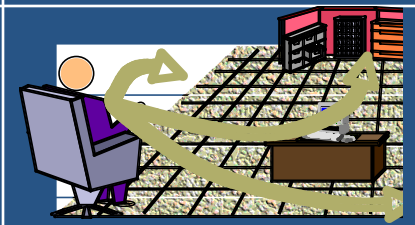
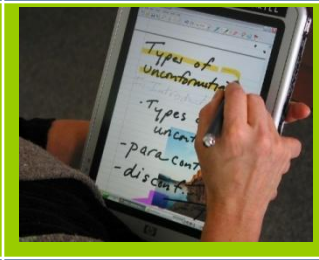
Stuart Card, Xerox, PARC

http://www.designinginteractions.com/interviews/StuCard

[Stu Card video from Moggridge Book]

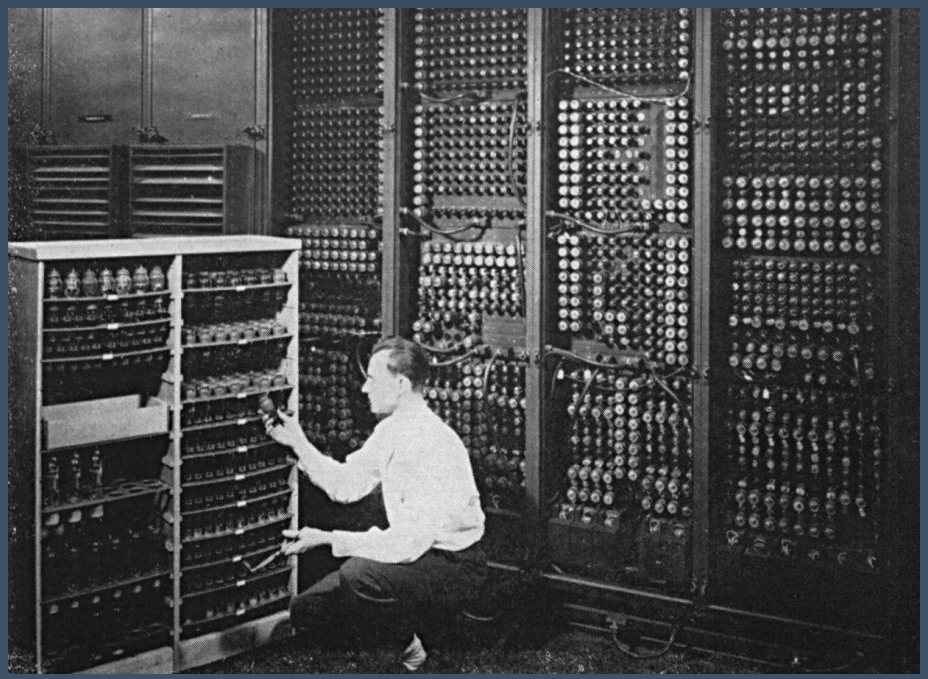
Source: Moggridge, Bill. Designing Interactions. MIT Press, 2007

TIME SCALE OF BEHAVIOR

10^7 (months) 10^6 (weeks) 10^5 (days)	**SOCIAL**	**Social Behavior**	
10^4 (hours) 10^3 10^2 (minutes)	**RATIONAL**	**Adaptive Behavior**	
10^1 10^0 (seconds) 10^{-1}	**COGNITIVE**	**Immediate Behavior**	
10^{-2} 10^{-3} (msec) 10^{-4}	**BIOLOGICAL**		

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

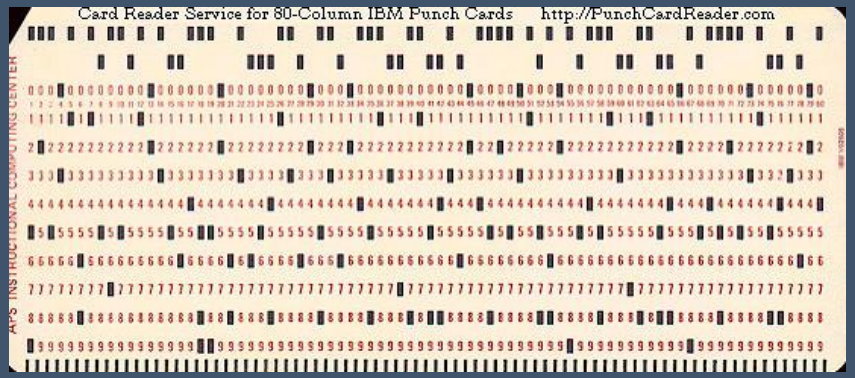
Ever wanted to have one?













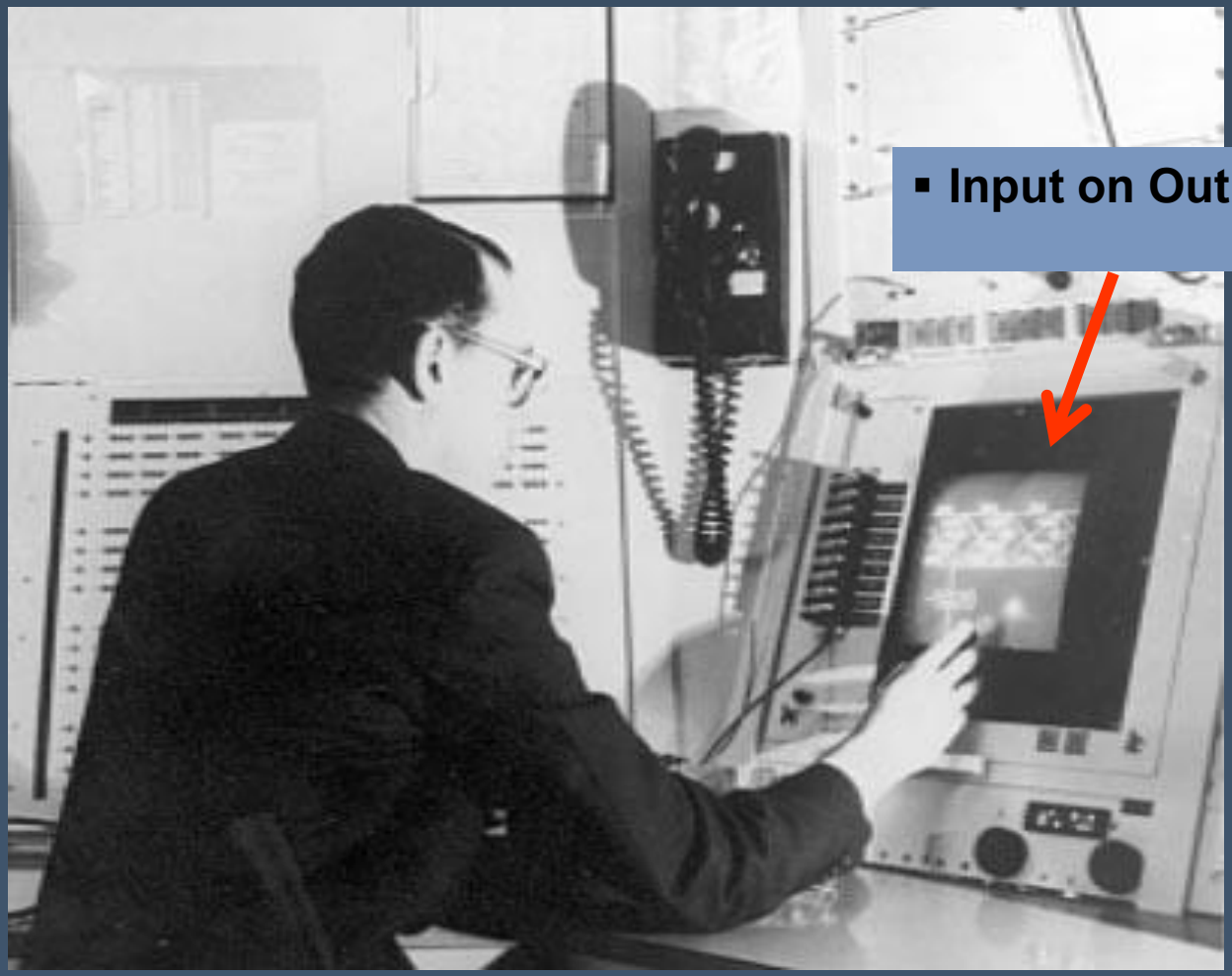
INTERACTIVE COMPUTING



Whirlwind (MIT-1951)

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

DIRECT MANIPULATION



Sketchpad (Sutherland, 1963)

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

EXAMPLE: POINTING DEVICES



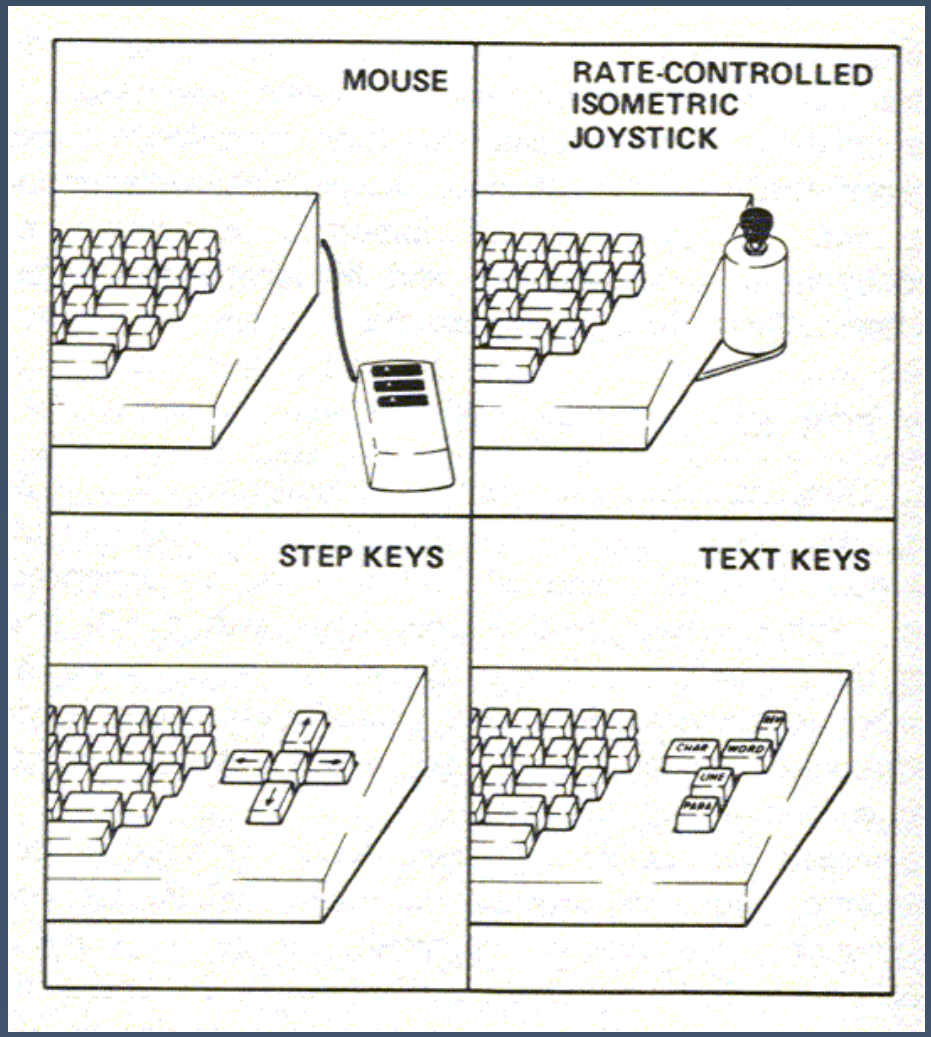
Mouse. Douglas Engelbart and Bill English (1964)

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

WHICH IS FASTEST?

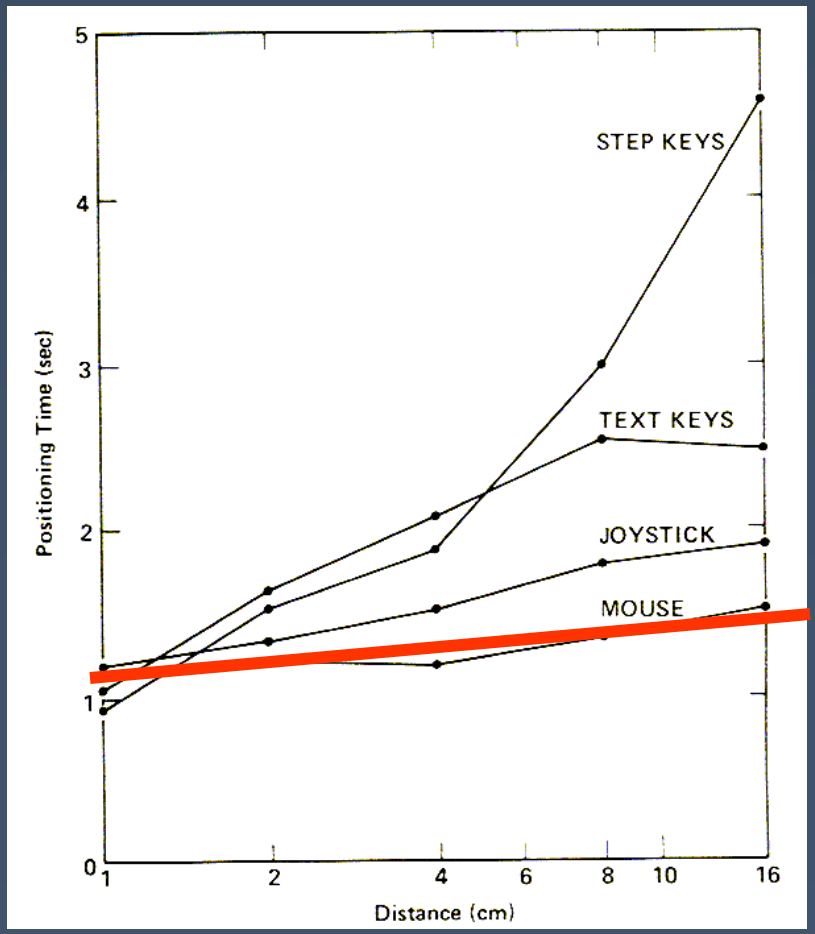


Engelbart



Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

EXPERIMENT: MICE ARE FASTEST



Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

WHY? (ENGINEERING ANALYSIS)



Why these results?

Time to position mouse proportional to Fitts' Index of Difficulty I_D.

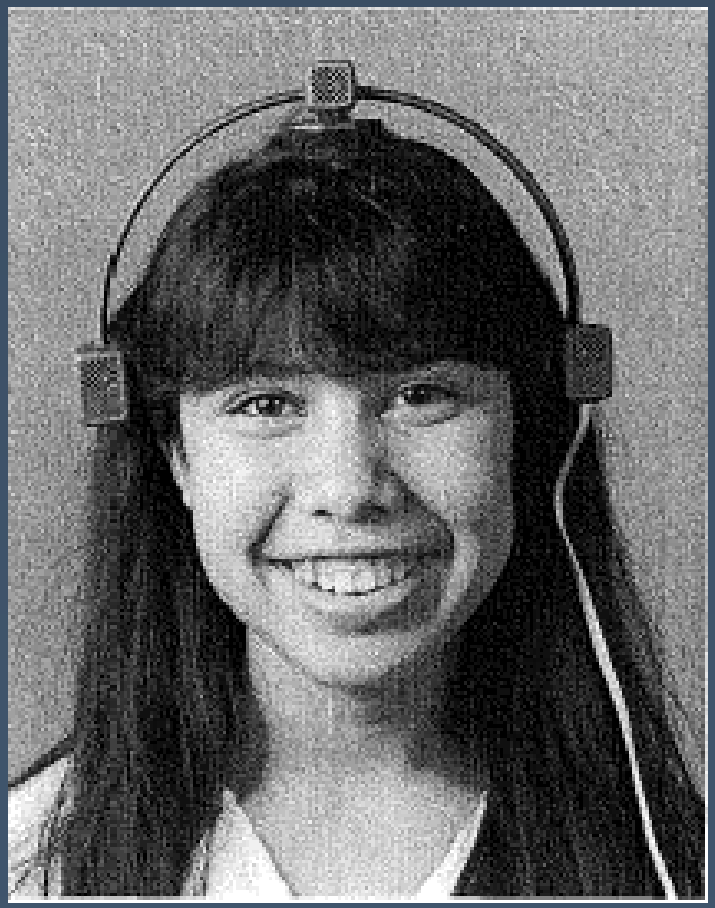
[i.e., how well can the muscles direct the input device]

Therefore, speed limit is in the eye-hand system, not the mouse.

Therefore, mouse is a near optimal device.

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

EXAMPLE: ALTERNATIVE DEVICES

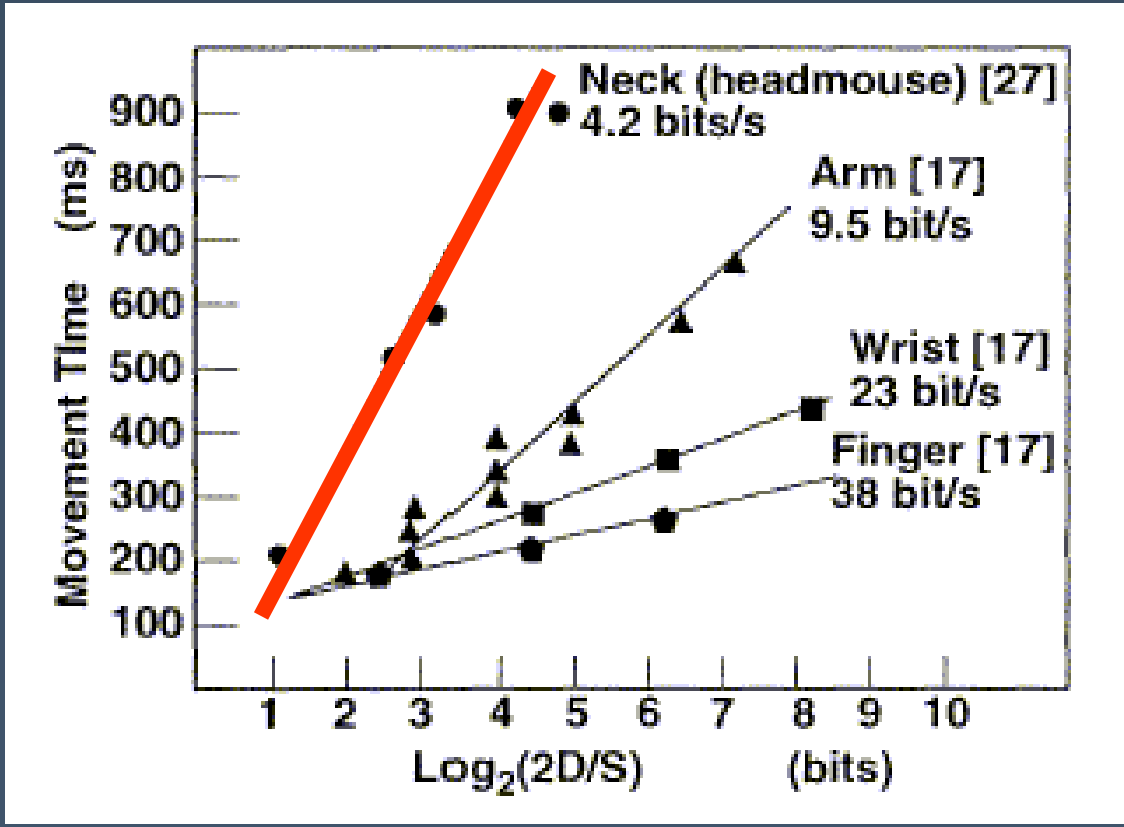




Headmouse: No chance to win

Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

PERFORMANCE OF HEADMOUSE



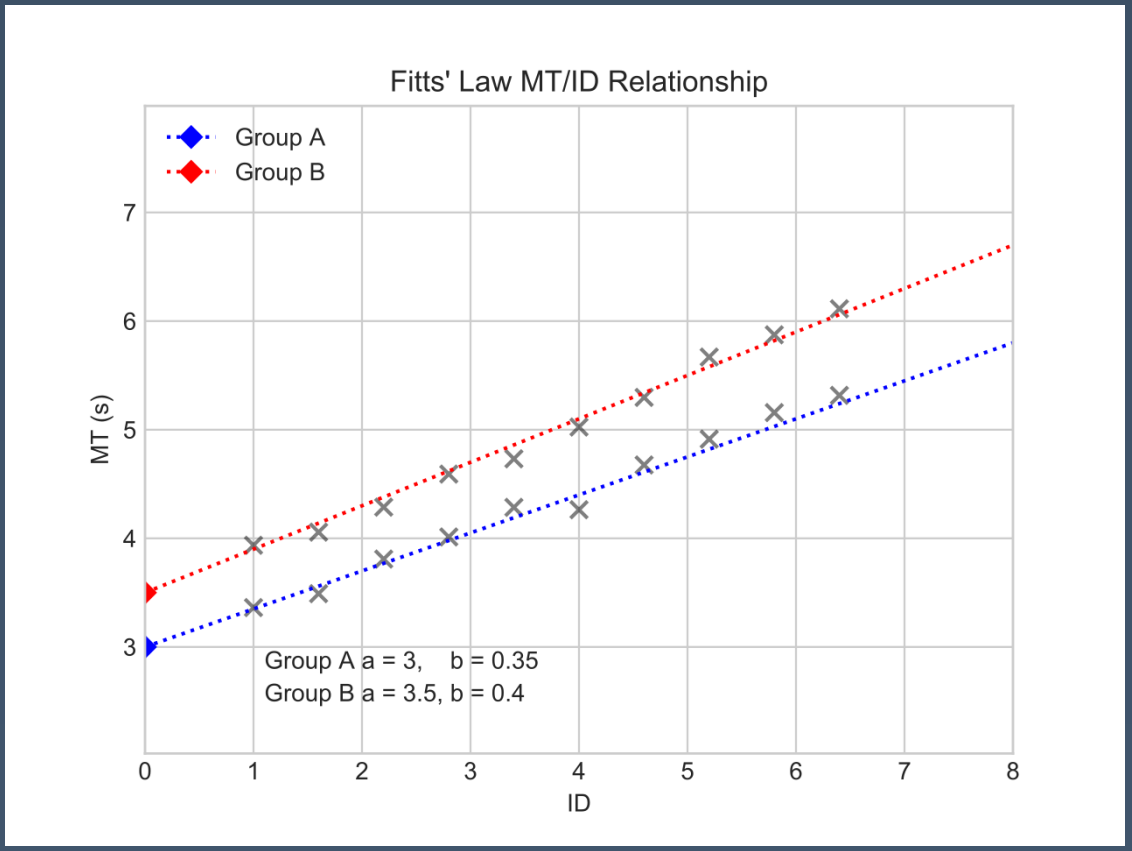
Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

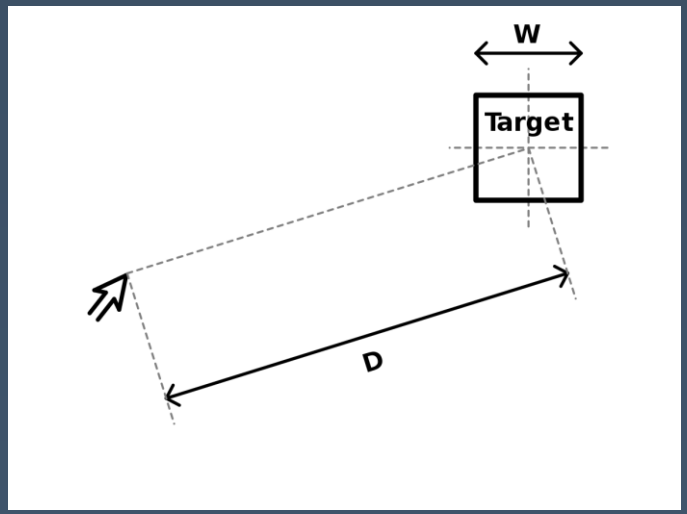
Principles of Operation

- Fitts' Law
  - Time T_{pos} to move the hand to target size W which is distance D away is given by:
    - $T_{pos} = a + b \log_2 (D/W + 1)$
- *a* and *b* are constants that depend on the choice of input device and are usually determined empirically by regression analysis. a defines the intersection on the y axis and is often interpreted as a delay. The b parameter is a slope and describes an acceleration. Both parameters show the linear dependency in Fitts's law.
  - summary
    - time to move the hand depends only on the ***relative precision*** required

Source: Landay, James. "Human Abilities". CS160 UC Berkeley.

Fitts' Law: *a*, *b* determination



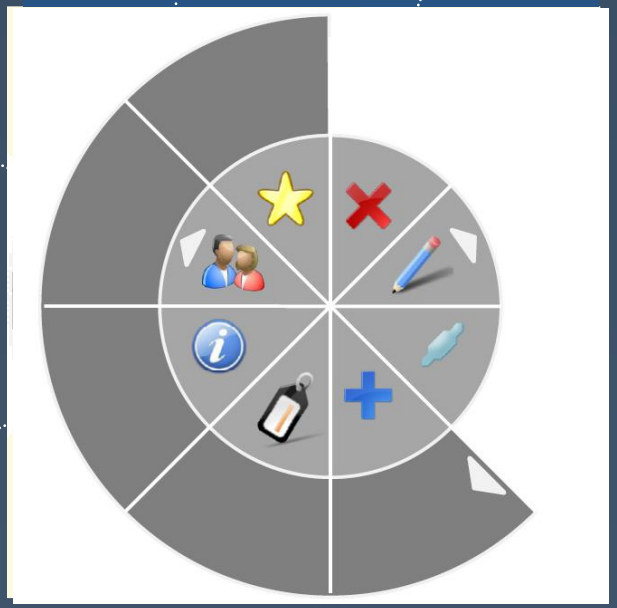


Fitts' Law Example

Pop-up Linear Menu

Today
Sunday
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday

Pop-up Pie Menu



- Which will be faster on average?
  - pie menu (bigger targets & less distance)

Source: Landay, James. "Human Abilities". CS160 UC Berkeley.

Fitt's Law in Windows vs Mac OS



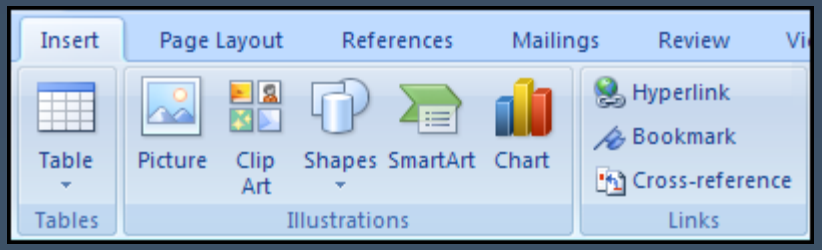
Windows 95: Missed by a pixel
Windows XP: Good to the last drop



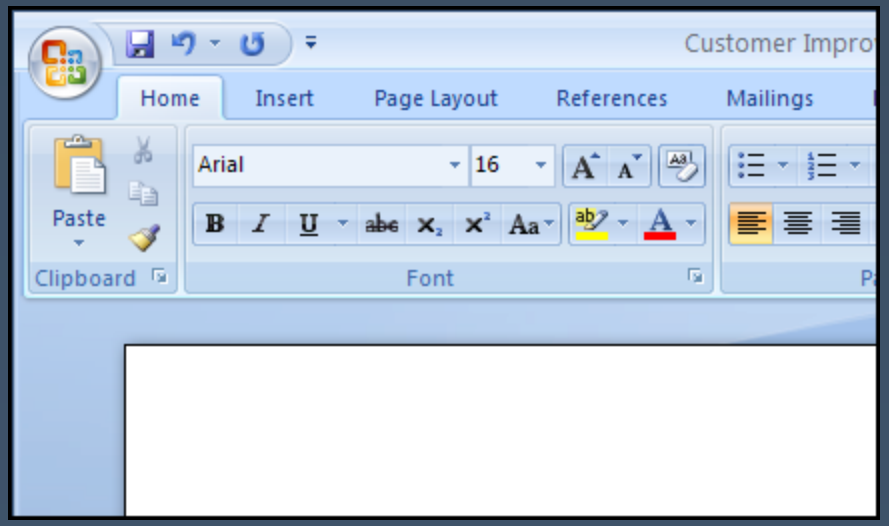
The Apple menu in Mac OS X v10.4 Tiger.

Source: Jensen Harris, An Office User Interface Blog : Giving You Fitts. Microsoft, 2007; Apple

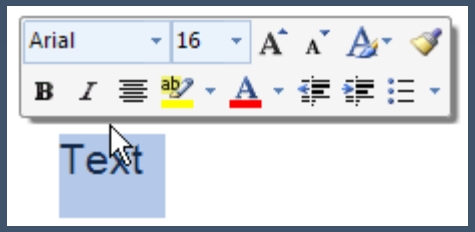
Fitt's Law in Microsoft Office 2007



Larger, labeled controls can be clicked more quickly



Magic Corner: Office Button in the upper-left corner



Mini Toolbar: Close to the cursor

Source: Jensen Harris, An Office User Interface Blog : Giving You Fitts. Microsoft, 2007.

Principles of Operation (cont.)

- Power Law of Practice
  - task time on the nth trial follows a power law
    - $T_n = T_1 n^{-a} + c$, where a = .4, c = limiting constant
    - i.e., you get faster the more times you do it!
    - applies to skilled behavior (sensory & motor)
    - does not apply to knowledge acquisition or quality

Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

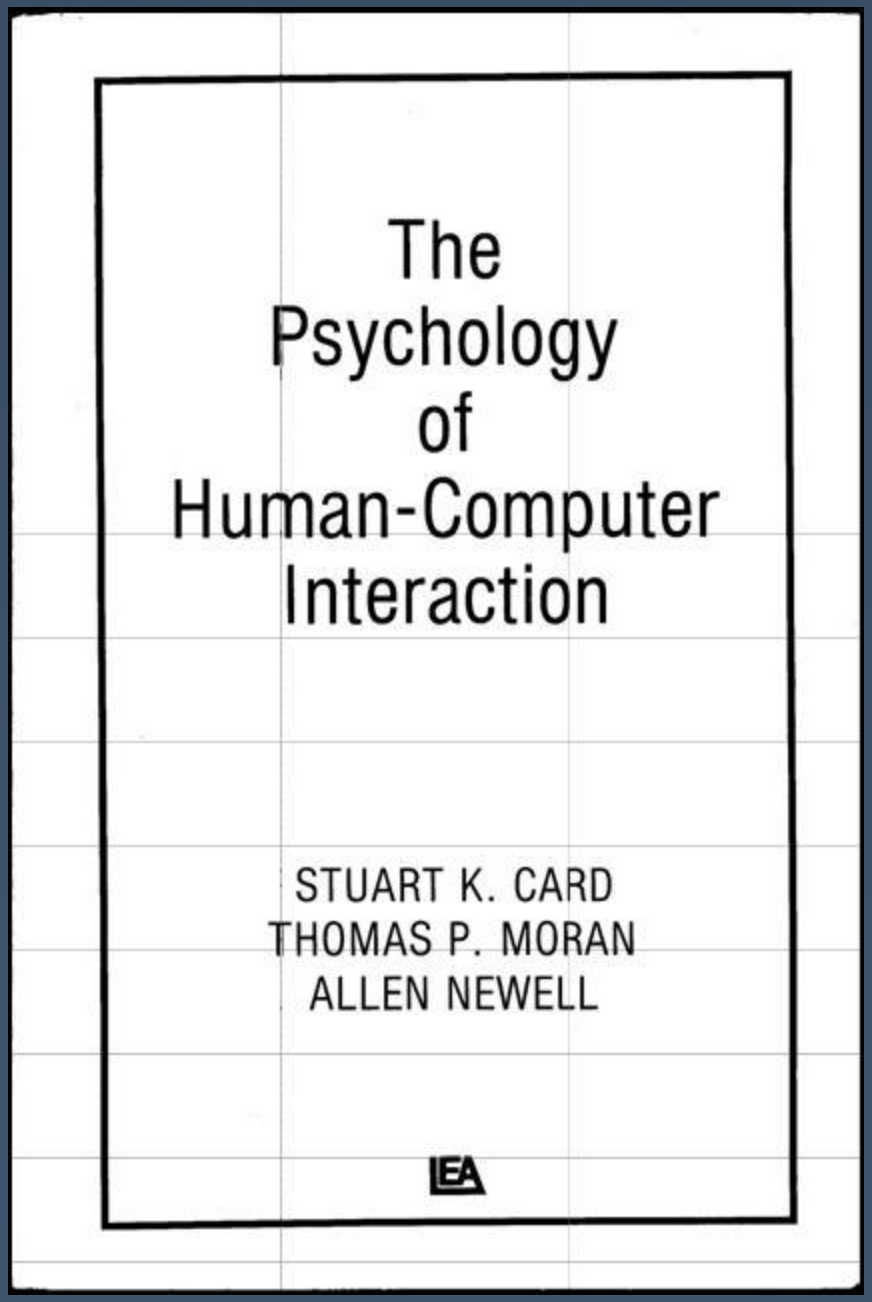
Implications for mobile design

- Nokia N95 interface designs?
- iPhone?



- What might happen to mobile device “inputs” in the future?

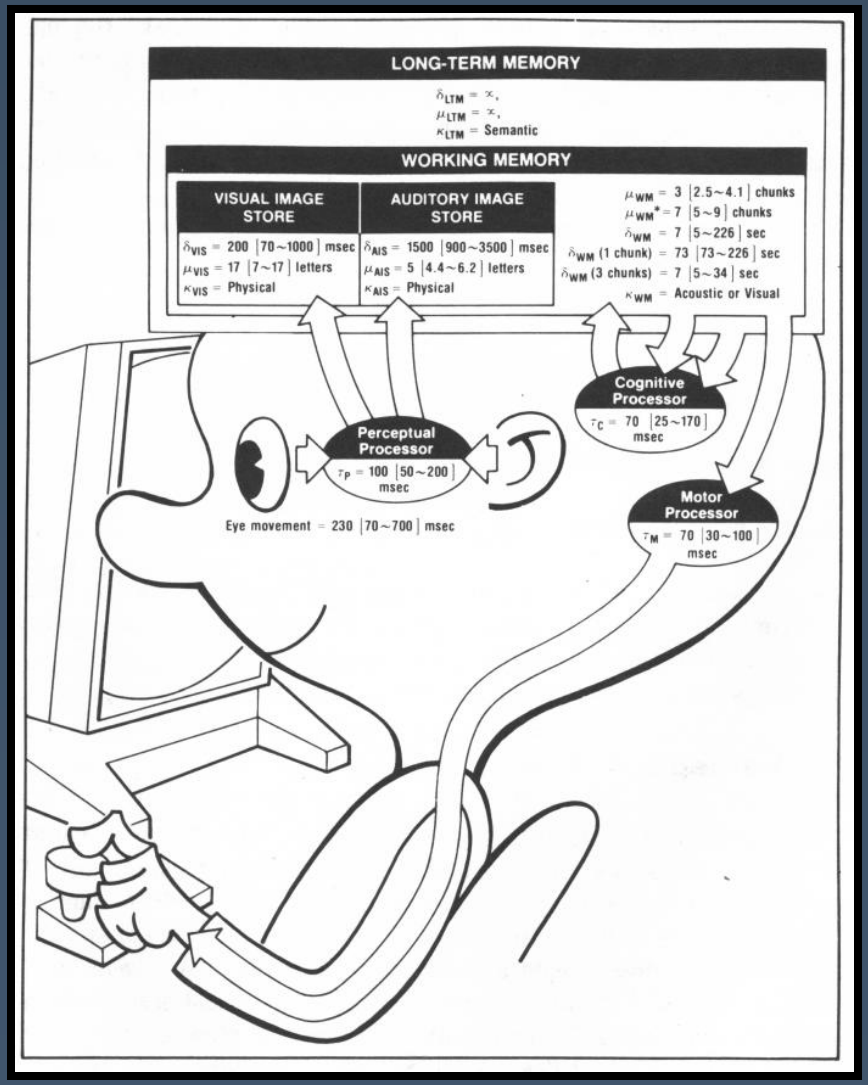
CMN



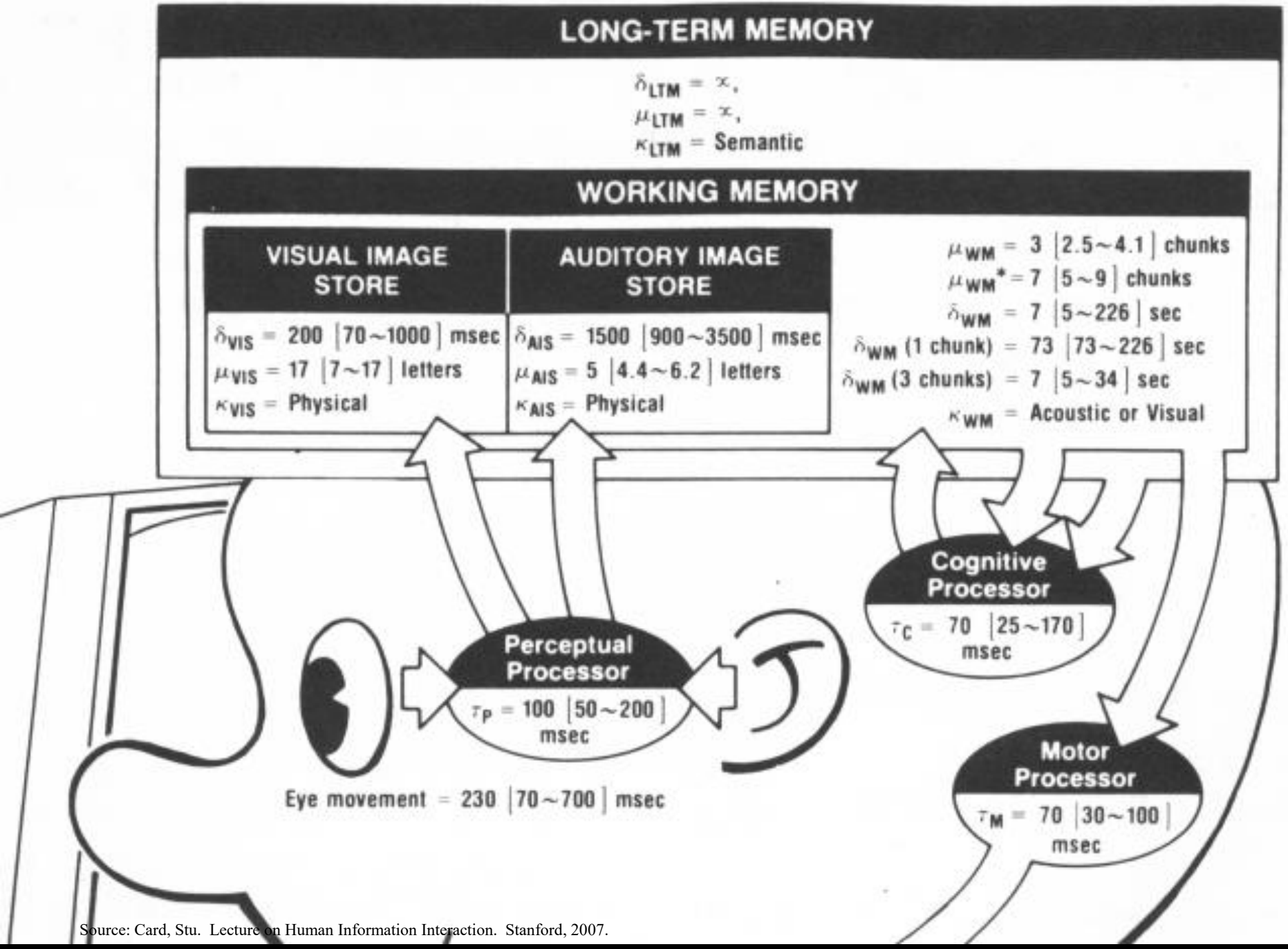
Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

MODEL HUMAN PROCESSOR

- Processors and Memories applied to human
- Used for routine cognitive skill [and learning and forgetting!]

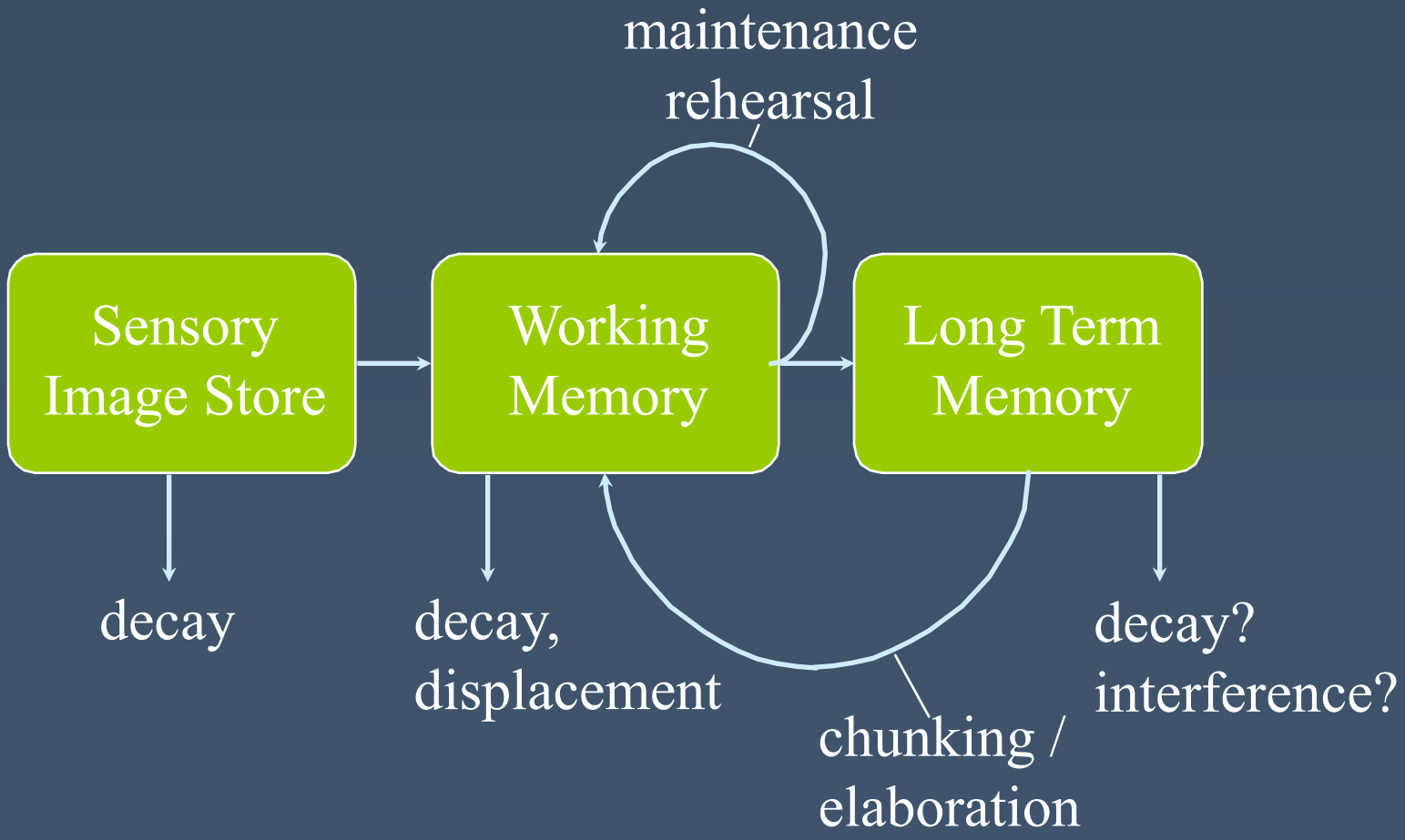


Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.



Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

Stage Theory



Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

Stage Theory

The ToA Stages (partial)

1. Decision
2. Planning
3. Translation
4. Execution
5. Segmentation

Stage Theory

Decision

Possible Problem:

- User fail to perform an action which they would have wished to do.

Example:

- Failing to save document before quiting.
- Failing to adjust settings before printing.

Suggested Solutions:

- Automatic saving - e.g. saving changes at specified time interval
- Visual interfaces - e.g. positioning of icons in declared temporary or permanent area.

Stage Theory

Planning

Possible Problem:

- Missing out on some steps during planning.
- Inserting text to the bottom or right-side of the page.

Example:

- Cut and paste procedure.
- Printing dialogue box.

Suggested Solutions:

- Executing a command which will lead to a string of steps to be performed.

Stage Theory

Translation

Possible Problem:

- Not being able to identify the command name for the desire action - e.g. graphs vs chart
- Not being able to remember the command itself, especially for command line driven applications.

Example:

- Display menu vs command line - e.g. MacOS vs Unix.

Suggested Solutions:

- On-line help to translate command names.
- Distinction of command options by length and letters.

Stage Theory

Execution

Possible Problem:

- Too much on screen movement for on-screen applications - e.g. MacOS

Example:

- Dragging an icon to the Wastebasket, especially on a big screen.
- Keyboard design.

Suggested Solutions:

- Alternate command on the menu to place items in the Wastebasket.
- Isolate dangerous keys from the frequently used ones.

Stage Theory

Segmentation

Possible Problem:

- Colour appearing differently in different machine.
- Dark font on dark background.
- Font style and size used in the wrong occasion.

Example:

- Programmers vs office clerks.
- Radio buttons and their labels.

Suggested Solutions:

- Dark fonts on light background.
- Use standardise colour and use single shade.
- Font style and size should be chosen with consideration of the ultimate users.

Stage Theory

- Working memory is small
  - temporary storage
    - decay
    - displacement
- Maintenance rehearsal
  - rote repetition
  - not enough to learn information well
- Answer to problem is organization

Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

MHP Principles of Operation

- Recognize-Act Cycle of the CP
  - on each cycle contents in WM initiate actions associatively linked to them in LTM
  - actions modify the contents of WM
- Discrimination Principle
  - retrieval is determined by candidates that exist in memory relative to retrieval cues
  - interference by strongly activated chunks

Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

Principles of Operation (cont.)

- Variable Cog. Processor Rate Principle
  - CP cycle time T_c is shorter when greater effort
  - induced by increased task demands/information
  - decreases with practice

Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

Implications for Designing from MHP

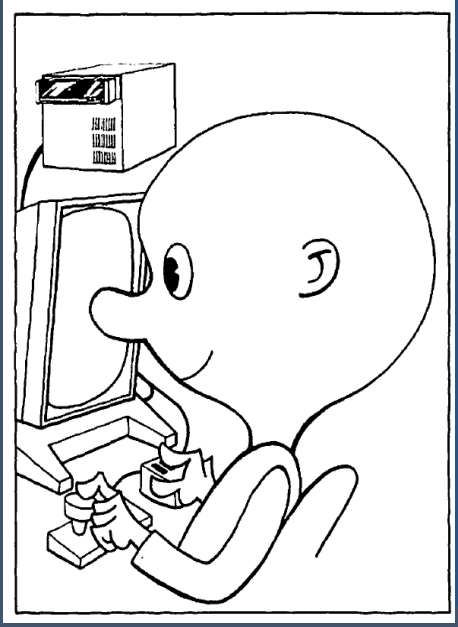
- Recognition over recall
- Relate interface to existing material
- Recode design in different ways
- Organize and link information
- Use visual imagery and auditory enhancements

Source: Landay, James. “Human Abilities”. CS160 UC Berkeley.

TASK ANALYSIS: GOMS (GOALS, OPERATORS, METHODS, SELECTION RULES)

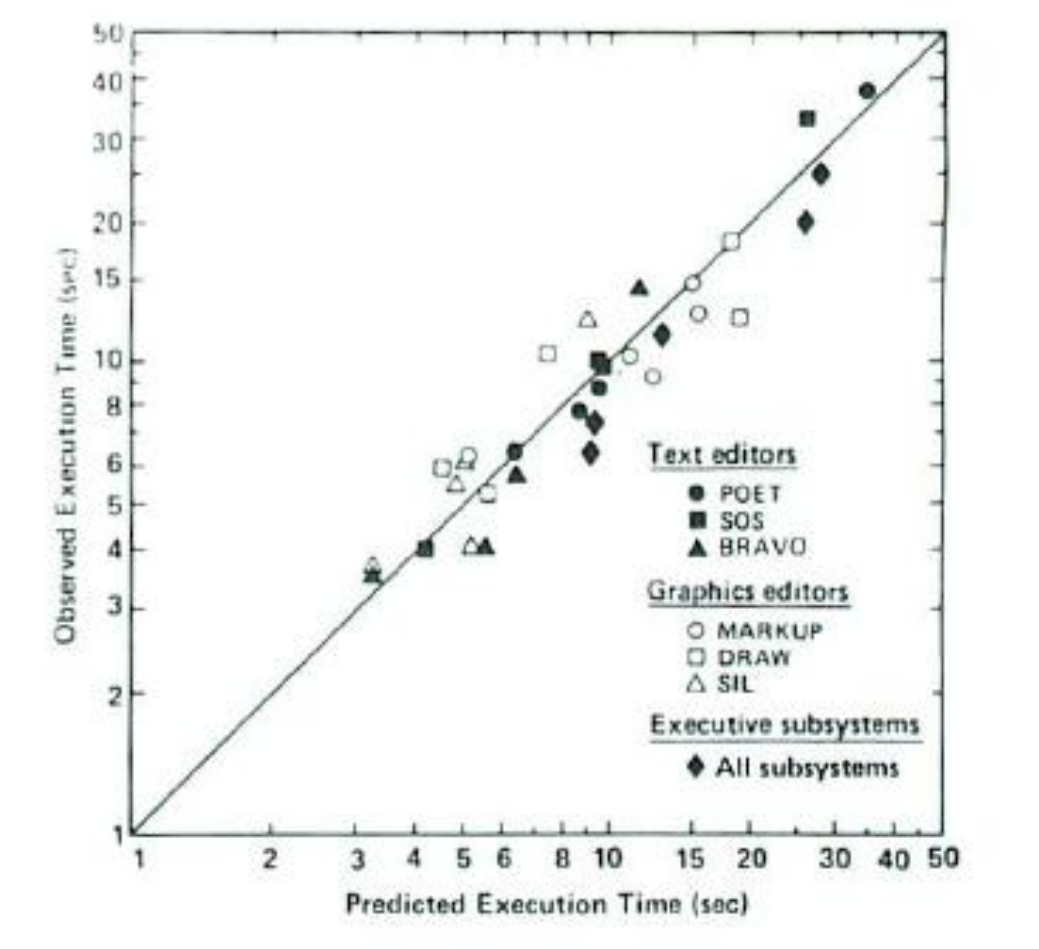
task analysis

GOAL: EDIT-MANUSCRIPT • repeat until done
GOAL: EDIT-UNIT-TASK
GOAL: ACQUIRE-UNIT-TASK • if not remembered
GET-NEXT-PAGE • if at end of page
GET-NEXT-TASK • if an edit task found
GOAL: EXECUTE-UNIT-TASK
GOAL: LOCATE-LINE • if task not on line
[select : USE-QS-METHOD
USE-LF-METHOD]
GOAL: MODIFY-TEXT
[select USE-S-COMMAND
USE-M-COMMAND]



Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

PREDICTS TIME WITHIN ABOUT 20%



Source: Card, Stu. Lecture on Human Information Interaction. Stanford, 2007.

GOMS Example: for Mac Finder

- Method for goal: delete a file.
- Step 1. Accomplish goal: drag file to trash.
- Step 2. Return with goal accomplished.
- Method for goal: move a file.
- Step 1. Accomplish goal: drag file to destination.
- Step 2. Return with goal accomplished.
- Method for goal: delete a directory.
- Step 1. Accomplish goal: drag directory to trash.
- Step 2. Return with goal accomplished.
- Method for goal: move a directory.
- Step 1. Accomplish goal: drag directory to destination.
- Step 2. Return with goal accomplished.



- Method for goal: drag item to destination.
- Step 1. Locate icon for item on screen.
- Step 2. Move cursor to item icon location.
- Step 3. Hold mouse button down.
- Step 4. Locate destination icon on screen.
- Step 5. Move cursor to destination icon.
- Step 6. Verify that destination icon is reverse-video.
- Step 7. Release mouse button.
- Step 8. Return with goal accomplished.

Source: Abowd, Gregory. CS 4753. Human Factors in Software Development. Georgia Tech.

Comparison: for DOS

- Method for goal: delete a file.
  - Step 1. Recall that command verb is "ERASE".
  - Step 2. Think of directory name and file name and retain as first filespec.
  - Step 4. Accomplish goal: enter and execute a command.
  - Step 6. Return with goal accomplished.
- Method for goal: move a file.
  - Step 1. Accomplish goal: copy a file.
  - Step 2. Accomplish goal: delete a file.
  - Step 3. Return with goal accomplished.
- Method for goal: copy a file.
  - Step 1. Recall that command verb is "COPY".
  - Step 2. Think of source directory name and file name and retain as first filespec.
  - Step 3. Think of destination directory name and file name and retain as second filespec.
  - Step 4. Accomplish goal: enter and execute a command.
  - Step 5. Return with goal accomplished.
- Method for goal: delete a directory.
  - Step 1. Accomplish goal: delete all files in the directory.
  - Step 2. Accomplish goal: remove a directory.
  - Step 3. Return with goal accomplished.
- Method for goal: delete all files in a directory.
  - Step 1. Recall that command verb is "ERASE".
  - Step 2. Think of directory name.
  - Step 3. Retain directory name and "*.*" as first filespec.
  - Step 4. Accomplish goal: enter and execute a command.
  - Step 5. Return with goal accomplished.
- Method for goal: remove a directory
  - Step 1. Recall that command verb is "RMDIR".
  - Step 2. Think of directory name and retain as first filespec.
  - Step 3. Accomplish goal: enter and execute a command.
  - Step 4. Return with goal accomplished.
- Method for goal: move a directory.
  - Step 1. Accomplish goal: copy a directory.
  - Step 2. Accomplish goal: delete a directory.
  - Step 3. Return with goal accomplished.
- Method for goal: copy a directory.
  - Step 1. Accomplish goal: create a directory.
  - Step 2. Accomplish goal: copy all the files in a directory.
  - Step 3. Return with goal accomplished.
- Method for goal: create a directory.
  - Step 1. Recall that command verb is "MKDIR".
  - Step 2. Think of directory name and retain as first filespec.
  - Step 3. Accomplish goal: enter and execute a command.
  - Step 4. Return with goal accomplished.
- Method for goal: copy all files in a directory.
  - Step 1. Recall that command verb is "COPY".
  - Step 2. Think of directory name.
  - Step 3. Retain directory name and "*.*" as first filespec.
  - Step 4. Think of destination directory name.
  - Step 5. Retain destination directory name and "*.*" as second filespec.
  - Step 6. Accomplish goal: enter and execute a command.
  - Step 7. Return with goal accomplished.



- Method for goal: enter and execute a command.
- *Entered with strings for a command verb and one or two filespecs.*
- Step 1. Type command verb.
- Step 2. Accomplish goal: enter first filespec.
- Step 3. Decide: If no second filespec, goto 5.
- Step 4. Accomplish goal: enter second filespec.
- Step 5. Verify command.
- Step 6. Type "<CR>".
- Step 7. Return with goal accomplished.
- Method for goal: enter a filespec.

- *Entered with directory name and file name strings.*
- Step 1. Type space.
- Step 2. Decide: If no directory name, goto 5.
- Step 3. Type "\".
- Step 4. Type directory name.
- Step 5. Decide: If no file name, return with goal accomplished.
- Step 6. Type file name.
- Step 7. Return with goal accomplished.

Source: Abowd, Gregory. CS 4753. Human Factors in Software Development. Georgia Tech.

Comparison

- Mac Finder: only 3 methods to accomplish these user goals, involving a total of only 18 steps.
- DOS requires 12 methods with a total of 68 steps.



- Consistency in Mac Finder
- A major value of a GOMS model is its ability to characterize, and even quantify, this property of *method consistency*.

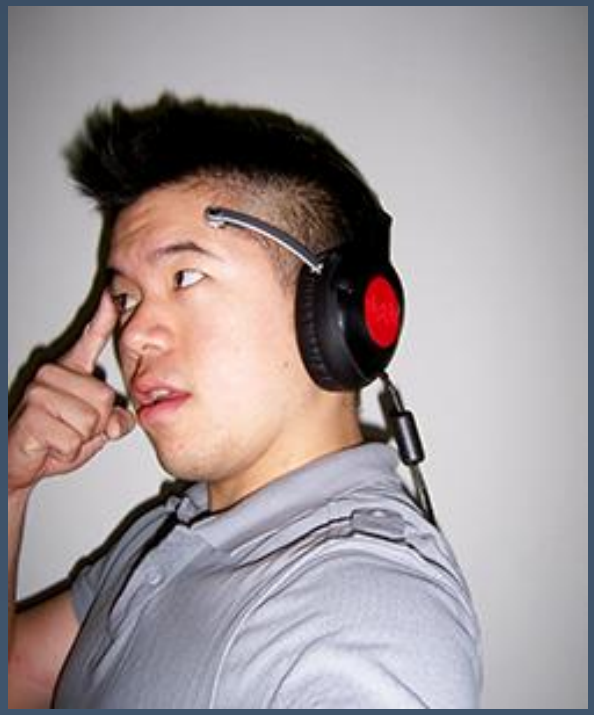
Source: Abowd, Gregory. CS 4753. Human Factors in Software Development. Georgia Tech.

Implications for interface design

- GOMS not often used formally
- But thinking through consistency of sub-tasks very useful!
- Good for comparing different systems

Eye to the Future: Brain Computer Interfaces

- Your brain might be your next videogame controller.
- http://www.youtube.com/watch?v=hQWBfCg91CU

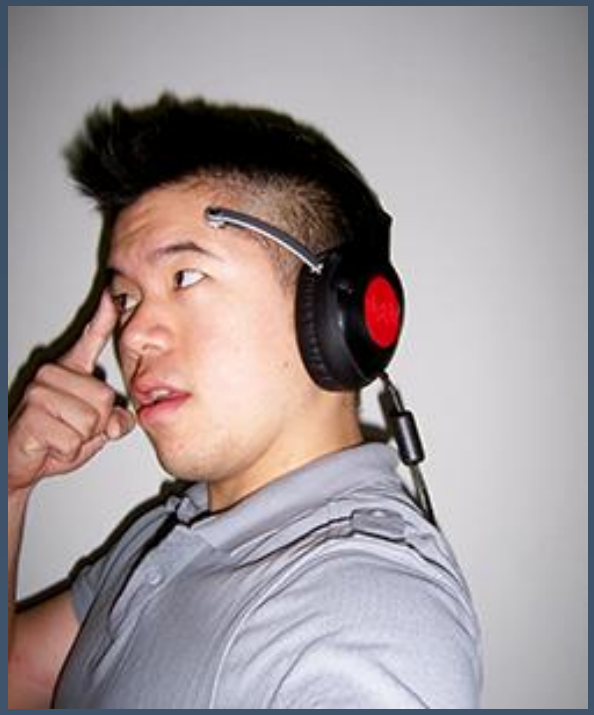


NeuroSky

Source: NeuroSky, “Direct Brain-to-Game Interface Worries Scientists”, Wired Magazine, 2007

Eye to the Future: Brain Computer Interfaces WARNING!

- … the devices sometimes force users to slow down their brain waves. Afterward, users have reported trouble focusing their attention.



NeuroSky

Source: NeuroSky, “Direct Brain-to-Game Interface Worries Scientists”, Wired Magazine, 2007